SMART PARKING SYSTEM

Java source code • Console-based academic project

Project description

This Java program manages a small parking area. It displays slot availability, assigns the first free slot to a vehicle, prevents the same vehicle from being parked twice, and allows a vehicle to exit. A fixed demonstration charge is displayed when a vehicle is removed; actual entry and exit time billing is not implemented.

How to run

```
1. Save the code as SmartParkingSystem.java
2. Open a terminal in that folder.
3. Compile: javac SmartParkingSystem.java
4. Run:     java SmartParkingSystem
```

Complete source code

```
import java.util.Scanner;

class ParkingSlot {
    private final int slotNumber;
    private boolean occupied;
    private String vehicleNumber;

    public ParkingSlot(int slotNumber) {
        this.slotNumber = slotNumber;
        this.occupied = false;
        this.vehicleNumber = "";
    }

    public int getSlotNumber() {
        return slotNumber;
    }

    public boolean isOccupied() {
        return occupied;
    }

    public String getVehicleNumber() {
        return vehicleNumber;
    }

    public void parkVehicle(String number) {
        occupied = true;
        vehicleNumber = number;
    }

    public void removeVehicle() {
        occupied = false;
        vehicleNumber = "";
    }
}

public class SmartParkingSystem {
    private static final int TOTAL_SLOTS = 10;
    private static final double HOURLY_RATE = 20.0;
    private static final ParkingSlot[] slots = new ParkingSlot[TOTAL_SLOTS];

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        for (int i = 0; i < TOTAL_SLOTS; i++) {
            slots[i] = new ParkingSlot(i + 1);
        }

        int choice;
        do {
            System.out.println("\\n===== SMART PARKING SYSTEM =====");
            System.out.println("1. View parking slots");
            System.out.println("2. Park a vehicle");
            System.out.println("3. Remove a vehicle");
            System.out.println("4. Check available slots");
            System.out.println("5. Exit");
            System.out.print("Enter your choice: ");

            while (!scanner.hasNextInt()) {
                System.out.print("Enter a valid number: ");
                scanner.next();
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        }
        choice = scanner.nextInt();
        scanner.nextLine();

        switch (choice) {
            case 1:
                displaySlots();
                break;
            case 2:
                parkVehicle(scanner);
                break;
            case 3:
                removeVehicle(scanner);
                break;
            case 4:
                System.out.println("Available slots: " + countAvailableSlots());
                break;
            case 5:
                System.out.println("Thank you for using Smart Parking System.");
                break;
            default:
                System.out.println("Invalid choice. Please try again.");
        }
    } while (choice != 5);

    scanner.close();
}

private static void displaySlots() {
    System.out.println("\\n--- PARKING SLOT STATUS ---");
    for (ParkingSlot slot : slots) {
        String status = slot.isOccupied()
                ? "Occupied (" + slot.getVehicleNumber() + ")"
                : "Available";
        System.out.println("Slot " + slot.getSlotNumber() + ": " + status);
    }
}

private static void parkVehicle(Scanner scanner) {
    int availableIndex = -1;

    for (int i = 0; i < slots.length; i++) {
        if (!slots[i].isOccupied()) {
            availableIndex = i;
            break;
        }
    }

    if (availableIndex == -1) {
        System.out.println("Sorry, the parking area is full.");
        return;
    }

    System.out.print("Enter vehicle number: ");
    String vehicleNumber = scanner.nextLine().trim();

    if (vehicleNumber.isEmpty()) {
        System.out.println("Vehicle number cannot be empty.");
        return;
    }

    for (ParkingSlot slot : slots) {
        if (slot.isOccupied()
                && slot.getVehicleNumber().equalsIgnoreCase(vehicleNumber)) {
            System.out.println("This vehicle is already parked.");
            return;
        }
    }

    slots[availableIndex].parkVehicle(vehicleNumber);
    System.out.println("Vehicle parked successfully.");
    System.out.println("Assigned slot: " + slots[availableIndex].getSlotNumber());
}

private static void removeVehicle(Scanner scanner) {
    System.out.print("Enter vehicle number to remove: ");
    String vehicleNumber = scanner.nextLine().trim();

    for (ParkingSlot slot : slots) {
        if (slot.isOccupied()
                && slot.getVehicleNumber().equalsIgnoreCase(vehicleNumber)) {
            slot.removeVehicle();
            System.out.println("Vehicle removed successfully.");
            System.out.printf("Demo parking charge: Rs. %.2f%n", HOURLY_RATE);
            return;
```

```
            }
        }

        System.out.println("Vehicle not found in the parking area.");
    }

    private static int countAvailableSlots() {
        int count = 0;
        for (ParkingSlot slot : slots) {
            if (!slot.isOccupied()) {
                count++;
            }
        }
        return count;
    }
}
```